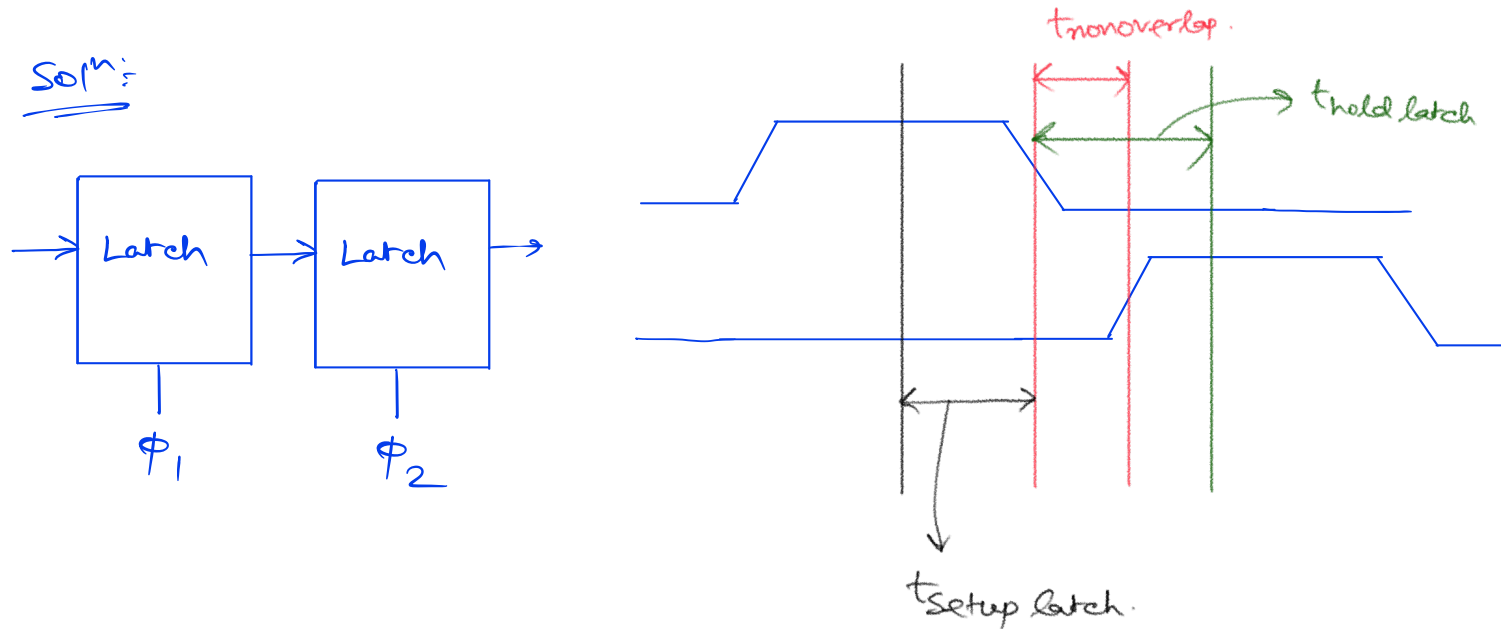


Question 5

Consider a flip-flop built from a pair of transparent latches using nonoverlapping clocks. Express the setup time, hold time, and clock-to-Q delay of the flip-flop in terms of the latch timing parameters and $t_{\text{nonoverlap}}$, relative to the rising edge of ϕ_1 .



$$\rightarrow t_{\text{Setup-F/F}} = t_{\text{Setup Latch}} + t_{\text{nonoverlap}}$$

$$\rightarrow t_{\text{hold-F/F}} = t_{\text{hold-latch}} - t_{\text{nonoverlap}}$$

$$\rightarrow t_{\text{CQ-F/F}} = t_{\text{CQ-Latch}}$$

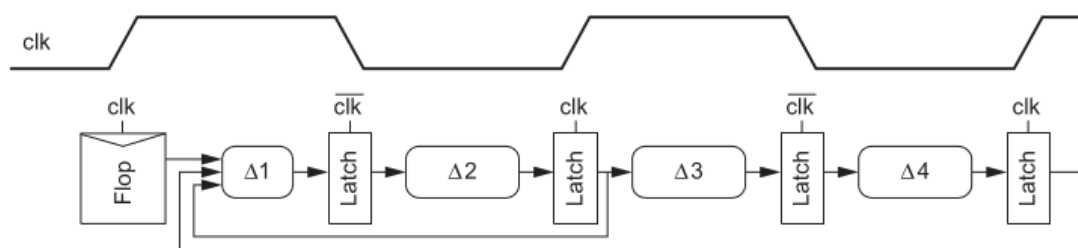
Question 6

Determine the minimum clock period at which the circuit in Figure below will operate correctly for each of the following logic delays. Assume there is zero clock skew and that the latch delays are accounted for in the propagation delay

(a) $\Delta_1 = 300$ ps; $\Delta_2 = 400$ ps; $\Delta_3 = 200$ ps; $\Delta_4 = 350$ ps

(b) $\Delta_1 = 300$ ps; $\Delta_2 = 400$ ps; $\Delta_3 = 400$ ps; $\Delta_4 = 550$ ps

(c) $\Delta_1 = 300$ ps; $\Delta_2 = 900$ ps; $\Delta_3 = 200$ ps; $\Delta_4 = 350$ ps



Soln:

path 1 = $\Delta_1 + \Delta_2$; path 2 = $\Delta_3 + \Delta_4$.

(a). path 1 = $300 + 400 = 700$ ps. ; path 2 = $200 + 350 = 550$ ps.

$t_c = \max(700 \text{ ps}, 550 \text{ ps}) = 700 \text{ ps}.$

(b) path 1 = $300 + 400 = 700$ ps ; path 2 = $400 + 550 = 950$ ps.

$t_c = \max(700 \text{ ps}, 950 \text{ ps}) = 950 \text{ ps}.$

(c). path 1 = $300 + 900 = 1200$ ps. ; path 2 = $200 + 350 = 550$ ps.

$t_c = \max(1200, 550) = 1200 \text{ ps}.$